Applied AI exam

Question 1)

a) Introduction to the Turing Test in AI (2 points)

The Turing Test, proposed by Alan Turing in 1950, evaluates a machine's ability to exhibit intelligent behavior indistinguishable from that of a human. In the test, a human judge interacts with both a machine and a human through text-based communication. If the judge cannot reliably differentiate between the two, the machine is said to have passed the test, demonstrating human-like intelligence.

Example: A chatbot like OpenAI's ChatGPT engaging in a conversation with a person. If the person cannot determine whether they are talking to a machine or another human, the chatbot passes the Turing Test.

b) Commonalities in Acting Rationally, Acting Humanly, Thinking Humanly, and Thinking Rationally in AI (5 points)

1. **Goal-Oriented Behavior**: All approaches aim to achieve specific objectives or solve problems efficiently.
2. **Knowledge Representation**: They require the ability to store and use knowledge to make decisions.
3. **Learning from Experience**: Adaptation and improvement over time through learning mechanisms are integral.
4. **Decision-Making Processes**: They involve reasoning and decision-making to choose the best course of action.

Question 2)

a) Key Aspects of an Intelligent Agent (5 points)

1. **Perception**: The ability to perceive the environment through sensors or data input.
2. **Reasoning**: The ability to process information and make decisions based on knowledge and logic.
3. **Learning**: The capability to learn from experiences and adapt over time.
4. **Autonomy**: The ability to operate independently without constant human intervention.
5. **Goal-Directed Behavior**: Acting in pursuit of specific objectives or tasks.

b) Factors to Consider When Designing an Intelligent Agent (4 points)

1. **Environment**: Understanding the nature of the environment (static or dynamic, deterministic or stochastic, discrete or continuous).
2. **Performance Measure**: Defining the metrics by which the agent's success will be evaluated.
3. **Agent Architecture**: Selecting an appropriate structure (reactive, deliberative, or hybrid) to suit the application.
4. **Adaptability**: Ensuring the agent can handle unforeseen changes and evolve with new data.

c) Designing a Utility-Based Intelligent Agent for a Real-World Application (6 points)

1. Define the Utility Function

- Identify and quantify goals or objectives of the application.
- Create a mathematical representation that assigns utility values to states or outcomes.

- Example: In a ride-hailing app, utility may depend on factors like minimizing waiting time, cost, and distance.

2. Manage Observation and Action Spaces

- Ensure the agent can observe relevant environmental variables (e.g., real-time traffic, user preferences).
- Define the action space (e.g., assigning drivers, pricing strategies).
- Use techniques like feature selection and dimensionality reduction to optimize these spaces.

3. Address Challenges of Adaptation

- Implement machine learning models for continuous improvement based on user feedback and environmental changes.
- Incorporate robust mechanisms to handle incomplete or noisy data.

4. Integrate into Real-World Environment

- Use APIs and scalable architectures to connect the agent with existing systems.
- Test in simulated environments before deploying in real-world settings.

5. Ensure Robustness and Scalability

- Design the system to handle high volumes of data and unexpected failures.
- Optimize computational resources to ensure performance under varying loads.

6. Evaluate and Iterate

- Regularly assess the agent's performance using the utility function and user satisfaction metrics.
- Use this feedback to refine the utility function, improve observation mechanisms, and expand action strategies.

Question 3) (10 Points)

Imagine a robot assistant designed to help you prepare a meal. Using the four AI paradigms, describe how the robot would function in each scenario:

Acting humanly: How would the robot behave if it were trying to imitate a human chef's actions in the kitchen?

Thinking humanly: If the robot's decision-making process were based on replicating human thought processes, how would it decide which ingredients to use and in what order to prepare the meal?

Acting rationally: How would the robot prepare the meal if its primary goal were to achieve the best result as efficiently as possible, without necessarily behaving like a human?

Thinking rationally: If the robot were using logical reasoning and algorithms to plan the meal preparation, what would its process be for selecting ingredients, timing, and cooking steps to ensure the best outcome?

1. Acting Humanly

If the robot were imitating a human chef's actions:

- It would mimic how a human chef chops vegetables, **measures ingredients**, and stirs food in the pot, possibly even engaging in small talk or offering commentary as it works.
- The **robot might mimic human imperfections**, such as hesitating to check a recipe or cleaning up spills.
- It could also use gestures that humans employ, like tasting food with a spoon to check seasoning.

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2. Thinking Humanly

If the robot's **decision-making** were based on replicating human thought processes:

- It would analyze recipes using cognitive processes similar to a human, like recalling past experiences (e.g., remembering a successful recipe it "learned").
- It might make decisions based on **sensory input**, like **detecting** a smell to judge whether food is burning.
- The robot could prioritize steps by reasoning, such as deciding to boil water first while chopping vegetables, because humans tend to multitask to save time.

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3. Acting Rationally

If the robot's **goal** were to prepare the meal efficiently and effectively:

- The robot would identify the fastest, most **energy-efficient way to prepare the meal**, such as using pre-chopped ingredients or setting timers for **optimal** cooking times.
- It would not mimic human behavior but instead **maximize** outcomes, such as using a sous-vide machine to achieve perfect cooking precision.
- It might skip **unnecessary actions like tasting food**, relying instead on pre-calibrated measurements and algorithms to ensure the correct flavor profile.

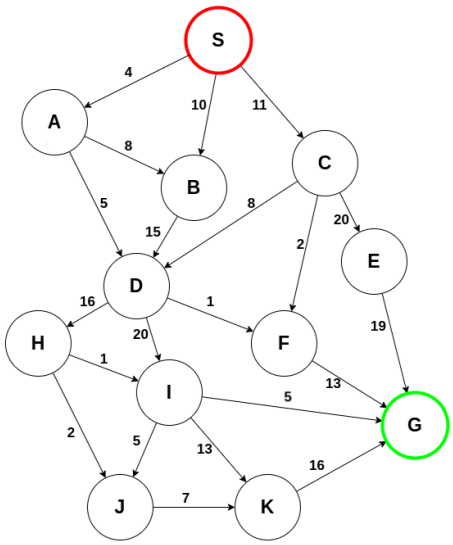
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4. Thinking Rationally

If the robot used logical reasoning and algorithms:

- The robot would evaluate the recipe as a **problem-solving task**, breaking it into **logical steps** like "select ingredients," "prepare ingredients," and "cook ingredients."
- It would use algorithms to calculate cooking times, adjust for portion sizes, and determine optimal sequencing (e.g., sauté onions first because they take longer than adding garlic).
- The robot would monitor **environmental factors** like temperature and adjust dynamically, ensuring each step leads to the best possible outcome without wasting resources or time.

Question 4)



Consider the graph and apply the following methods to find the shortest path from S to G:

a) Breadth-First Search (BFS). Show the solution in details. (4 points)

b) The A* algorithm. Show the solution in details. (6 points).

Solution:

a)

Breadth-First Search explores the graph level by level, starting from the source node S. It visits all immediate neighbors before moving to the next depth level. Assuming the goal is to find the shortest path from S to G:

Steps for BFS:

1. Start at node S and enqueue it in the frontier (queue): **Queue = [S]**.
2. Dequeue S and enqueue all its unvisited neighbors: **Queue = [A, B, C]**.
3. Dequeue A and enqueue its unvisited neighbors: **Queue = [B, C, D]**.
4. Dequeue B and enqueue its unvisited neighbors: **Queue = [C, D, E]**.
5. Dequeue C and enqueue its unvisited neighbors: **Queue = [D, E, F]**.
6. Dequeue D and enqueue its unvisited neighbors: **Queue = [E, F, I]**.
7. Dequeue E and enqueue its unvisited neighbors: **Queue = [F, I, G]**.
8. Dequeue F, add its neighbors if unvisited: **Queue = [I, G]**.
9. Dequeue I, add its neighbors if unvisited: **Queue = [G]**.
10. Dequeue G (Goal reached).

Solution Path: Trace back from G to S via the shortest route using BFS exploration order:

- $S \rightarrow C \rightarrow F \rightarrow G$.

b) A Algorithm*

The A* algorithm uses a combination of the actual cost $g(n)$ to reach a node and a heuristic $h(n)$ to estimate the cost from the node to the goal. Let $f(n) = g(n) + h(n)$.

Assumptions:

- Assume $h(n)$ values (heuristics) for each node are provided or inferred from the graph. If not explicitly available, please provide a heuristic table.

Steps for A*:

1. Start at node S with $f(S) = g(S) + h(S)$.
2. Expand the node with the lowest $f(n)$ value.
3. Update $g(n)$ for all neighbors of the expanded node.
4. Repeat until the goal node G is reached.

heuristic values $h(n)$ to each node based on an estimation of their distance to G:

Node	Heuristic $h(n)$
S	15
A	14
B	12
C	10
D	8
E	6
F	5
G	0
H	10
I	7
J	12
K	6

1. **Start at S:**
   - $g(S) = 0$, $h(S) = 15$, $f(S) = 0 + 15 = 15$.
   - Expand S and calculate $f(n)$ for its neighbors A, B, and C:
     - $f(A) = g(A) + h(A) = (0 + 4) + 14 = 18$.
     - $f(B) = g(B) + h(B) = (0 + 10) + 12 = 22$.
     - $f(C) = g(C) + h(C) = (0 + 11) + 10 = 21$.
2. **Expand A:**
   - $g(A) = 4$.
   - Neighbors of A: B, D.
     - $f(B) = g(B) + h(B) = (4 + 8) + 12 = 24$ (already in the frontier, no update).
     - $f(D) = g(D) + h(D) = (4 + 5) + 8 = 17$.
3. **Expand D:**
   - $g(D) = 9$.
   - Neighbors of D: B, F, I.
     - $f(B) = g(B) + h(B) = (9 + 15) + 12 = 36$ (no update as it's worse).
     - $f(F) = g(F) + h(F) = (9 + 1) + 5 = 15$.
     - $f(I) = g(I) + h(I) = (9 + 20) + 7 = 36$.
4. **Expand F:**
   - $g(F) = 10$.
   - Neighbors of F: G.
     - $f(G) = g(G) + h(G) = (10 + 5) + 0 = 15$.

Final Solution:

The path is $S \to A \to D \to F \to G$.

Costs for the Path:

- $S \to A$: Cost = 4.
- $A \to D$: Cost = 5.
- $D \to F$: Cost = 1.
- $F \to G$: Cost = 5.

Total Cost: $g(G) = 4 + 5 + 1 + 5 = 15$.

Path: $S \to A \to D \to F \to G$.

Question 5)

a) *What are the properties of the Environment of the Wumpus Problem? (3 points)*

b) ***Question****: Given the grid layout for a Wumpus World, what steps should the agent take to collect the gold and avoid the Wumpus? Describe the agent's movements and any decisions made based on perceptual clues. (10 points)*

Solution:

a)

1. **Discrete**: The environment is divided into a grid-based layout, where each cell represents a specific location.
2. **Partially Observable**: The agent can perceive only its immediate surroundings (e.g., stench, breeze, glitter) and must infer the rest of the environment.
3. **Deterministic**: The effects of the agent's actions (e.g., moving, shooting, grabbing) are predictable, and there is no randomness in the outcomes.
4. **Static**: The environment does not change while the agent is deliberating; the Wumpus and hazards remain in fixed positions.
5. **Episodic/Sequential**: Depending on how the agent interacts, it can behave in a sequential manner where actions and decisions build upon one another.

b) The agent uses sensing, detecting, decision-making, and memory throughout the process.

Step 1: Initial Position $(1,1)$

- **Sensing**: Perceptions: **No stench**, **No breeze**.
- **Detecting**:
  - The absence of stench confirms there is no Wumpus in adjacent cells ($(1,2), (2,1)$).
  - The absence of breeze confirms there are no pits in adjacent cells ($(1,2), (2,1)$).
- **Memory**: Mark $(1,1)$ as visited and safe.
- **Decision-Making**: Move to $(1,2)$ to explore further.

Step 2: Move to $(1,2)$

- **Sensing**: Perceptions: **Stench detected**, **No breeze**.
- **Detecting**:
  - **Stench**: Indicates the Wumpus is in one of the adjacent cells: $(1,3)$ or $(2,2)$.
  - **No breeze**: Confirms no pits in $(1,1), (1,3), (2,2)$.
- **Memory**: Mark $(1,2)$ as visited and safe. Remember $(1,3)$ as a potential Wumpus location.
- **Decision-Making**: Move to $(2,2)$, as it is unexplored

Step 3: Move to $(2, 2)$

- **Sensing**: Perceptions: **No stench**, **No breeze**.
- **Detecting**:
  - The absence of stench confirms there is no Wumpus in adjacent cells ($(2, 3), (3, 2)$).
  - The absence of breeze confirms there are no pits in adjacent cells ($(2, 3), (3, 2)$).
- **Memory**: Mark $(2, 2)$ as visited and safe. Confirm $(2, 3)$ is safe to explore.
- **Decision-Making**: Move to $(2, 3)$ to gather more information.

Step 4: Move to $(2, 3)$

- **Sensing**: Perceptions: **Breeze detected**, **Stench detected**.
- **Detecting**:
  - **Breeze**: Confirms the presence of a pit in one of the adjacent cells: $(3, 3)$ or $(2, 4)$.
  - **Stench**: Confirms the Wumpus is in $(1, 3)$.
- **Memory**: Mark $(2, 3)$ as visited and note it is adjacent to a pit.
- **Decision-Making**:

Gold is detected at (2,3) by the intelligent agent and the goal is achieved.

Question 6)

Convert x_1, x_2, x_3, x_4**:**

- $x_1 = 0110\,1001\,0011$
  - $a = 0110$: 6
  - $b = 1001$: 9
  - $c = 0011$: 3
  - Fitness: $f(x) = (a - b)^2 + c = (6 - 9)^2 + 3 = 9 + 3 = 12$
- $x_2 = 1011\,0110\,0100$
  - $a = 1011$: 11
  - $b = 0110$: 6
  - $c = 0100$: 4
  - Fitness: $f(x) = (a - b)^2 + c = (11 - 6)^2 + 4 = 25 + 4 = 29$

- $x_3 = 0010\,1110\,0111$
  - $a = 0010$: 2
  - $b = 1110$: 14
  - $c = 0111$: 7
  - Fitness: $f(x) = (a - b)^2 + c = (2 - 14)^2 + 7 = 144 + 7 = 151$
- $x_4 = 1100\,0101\,1000$
  - $a = 1100$: 12
  - $b = 0101$: 5
  - $c = 1000$: 8
  - Fitness: $f(x) = (a - b)^2 + c = (12 - 5)^2 + 8 = 49 + 8 = 57$

Arrange Chromosomes by Fitness:

1. $x_3 : f(x) = 151$ (Fittest)
2. $x_4 : f(x) = 57$
3. $x_2 : f(x) = 29$
4. $x_1 : f(x) = 12$ (Least Fit)

b)

Step 1: Select Second and Third Fittest Chromosomes

- Second Fittest: $x_4 = 1100\,0101\,1000$
- Third Fittest: $x_2 = 1011\,0110\,0100$

Step 2: Perform Crossover at Midpoint

- Midpoint = 6 bits (chromosome length: 12 bits).
- Split x_4: $1100\,01|01\,1000$
- Split x_2: $1011\,01|10\,0100$

Crossover Result:

- Offspring 1: $1100\,01|10\,0100 = 1100\,0110\,0100$
- Offspring 2: $1011\,01|01\,1000 = 1011\,0101\,1000$

c)

Convert Offspring 1:

- $1100\,0110\,0100$:
  - $a = 1100 = 12$
  - $b = 0110 = 6$
  - $c = 0100 = 4$
  - Fitness: $f(x) = (a - b)^2 + c = (12 - 6)^2 + 4 = 36 + 4 = 40$

Convert Offspring 2:

- $1011\,0101\,1000$:
  - $a = 1011 = 11$
  - $b = 0101 = 5$
  - $c = 1000 = 8$
  - Fitness: $f(x) = (a - b)^2 + c = (11 - 5)^2 + 8 = 36 + 8 = 44$

Compare Overall Fitness:

Original Population Fitness = $12 + 29 + 151 + 57 = 249$
New Population Fitness (with offspring replacing x_4 and x_2) = $12 + 40 + 44 + 151 = 247$

Observation: Overall fitness slightly decreased.

d)

- $a = 1111$ (in binary, 15 in decimal),
- $b = 0000$ (in binary, 0 in decimal),
- $c = 1111$ (in binary, 15 in decimal),
- Fitness function:

$$f(x) = (a - b)^2 + c$$

Calculation:

1. Compute $a - b$:

$$a - b = 15 - 0 = 15$$

2. Compute $(a - b)^2$:

$$(15)^2 = 225$$

3. Add c:

$$f(x) = 225 + 15 = 240$$

Result:

- The fitness value $f(x) = 240$.

e)

The genetic algorithm's convergence depends on:

Population Diversity:

The initial population shows diversity, but if future generations focus only on fitter individuals, diversity could decrease, risking premature convergence.

Crossover Points:

A single crossover point preserves much of the genetic information but may not explore new combinations effectively.

Mutation:

Mutation introduces variability and prevents stagnation. If no mutations are applied, the algorithm may not escape local optima.

Analysis:

The GA is likely to converge to a good solution if mutation is introduced to maintain diversity and prevent premature convergence. Without mutation, the algorithm risks getting stuck in suboptimal solutions due to limited exploration of the solution space.

Question 7) (13 points)

Solution:

(a) Linear Regression Model

Using the training data ($x = [1, 2, 3, 4, 5, 6, 7, 8]$, $y = [3, 5, 7, 6, 9, 11, 12, 14]$):

- **Intercept (b)**: $b = 1.5714$
- **Slope (m)**: $m = 1.5119$

The regression equation becomes:

$$y = 1.5714 + 1.5119 \cdot x$$

(b) Predictions

Using the regression model, the predicted y-values for the last two points ($x = [9, 10]$) are:

$$\text{For } x = 9: \; y_{\text{pred}} = 1.5714 + 1.5119 \cdot 9 = 15.1786$$

$$\text{For } x = 10: \; y_{\text{pred}} = 1.5714 + 1.5119 \cdot 10 = 16.6905$$

Predicted values:
$y_{\text{pred}} = [15.1786, 16.6905]$

Mean Squared Error (MSE)

The actual y-values for the last two points are $y_{\text{test}} = [15, 17]$. The MSE is calculated as:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_{\text{true}} - y_{\text{pred}})^2$$

$$\text{MSE} = 0.0638$$

Question 8)

a) *You have a dataset of house prices with a range of values from $150,000 to $1,500,000. Perform min-max normalization on the dataset to scale the values to a range between 1 and 5. Calculate the normalized value for a house priced at $800,000 using the min-max normalization formula. (3 points)*

b) *Explain a potential limitation of min-max normalization. What would happen to the normalized values if a new house price of $1,600,000 were added to the dataset? (6 points)*

c) *In a dataset of student exam scores, the mean score is 75 with a standard deviation of 10. Apply z-score normalization to a student's score of 85. Calculate the z-score for this score and determine how many standard deviations it is above or below the mean. (3 points)*

Solution:

(a) Min-Max Normalization

The min-max normalization formula is:

$$\text{Normalized Value} = \text{New Min} + \frac{(\text{Value} - \text{Min})}{\text{Max} - \text{Min}} \cdot (\text{New Max} - \text{New Min})$$

Given:

- Dataset range: $\text{Min} = 150,000$, $\text{Max} = 1,500,000$
- New range: $[1, 5]$
- House price: $800,000$

$$\text{Normalized Value} = 1 + \frac{(800,000 - 150,000)}{1,500,000 - 150,000} \cdot (5 - 1)$$

$$\text{Normalized Value} = 1 + \frac{650,000}{1,350,000} \cdot 4$$

$$\text{Normalized Value} = 1 + 0.4815 \cdot 4 = 2.9259$$

(b) Limitation of Min-Max Normalization

Limitation: Min-max normalization is sensitive to outliers. If the range of the dataset changes, such as when new extreme values are added, all previously normalized values must be recalculated.

Example: If a new house price of $1,600,000$ is added:

- The dataset maximum becomes $1,600,000$.
- The normalized values for all prices would change because the new maximum alters the scaling factor.

For example, the normalization of $800,000$ would become:

$$\text{Normalized Value} = 1 + \frac{(800,000 - 150,000)}{1,600,000 - 150,000} \cdot (5 - 1)$$

Normalized Value=2.7931

(c) Z-Score Normalization

The z-score normalization formula is:

$$z = \frac{\text{Value} - \text{Mean}}{\text{Standard Deviation}}$$

Given:

- Mean = 75
- Standard deviation = 10
- Student score = 85

$$z = \frac{85 - 75}{10} = \frac{10}{10} = 1.0$$

Question 9) (10 points)

Write down four ethical considerations surrounding AI, and why is it essential to address them in AI development and deployment? (10 points)

Solution:

1. **Bias and Fairness**
   - **Explanation**: AI systems can inherit or amplify biases present in the training data, leading to unfair or discriminatory outcomes. For example, biased hiring algorithms may disadvantage certain demographic groups.
   - **Why Address It**: Ensuring fairness in AI systems helps prevent discrimination, ensures equal opportunities, and builds trust in AI solutions.
2. **Privacy and Data Security**
   - **Explanation**: AI systems often require access to large amounts of personal data. Mishandling this data can lead to privacy breaches and misuse of sensitive information.
   - **Why Address It**: Protecting individuals' privacy and securing their data is crucial to maintain user trust, comply with regulations (like GDPR), and prevent misuse or harm.

3. **Transparency and Accountability**
    - **Explanation**: AI systems can operate as "black boxes," making decisions that are difficult to interpret or explain. This lack of transparency raises concerns about accountability, especially when mistakes or harm occur.
    - **Why Address It**: Transparency and clear accountability ensure that humans remain responsible for AI decisions, fostering trust and enabling users to challenge or audit outcomes when necessary.
4. **Job Displacement and Economic Impact**
    - **Explanation**: The automation of tasks by AI may displace workers, leading to job losses in certain industries while creating economic disparities.
    - **Why Address It**: Proactively addressing job displacement helps create strategies for workforce retraining and ensures that the benefits of AI are distributed equitably across society.